

Has the euro weakened?

Analysis report — EUR/USD and EUR/CHF

Data Analysis with Python · Digital Learning Hub Luxembourg

	Your answer
Your name	Sepideh Maleki Roudposhti
Date	9/17/2026
Data source	European Central Bank — euro reference rates
Date range you loaded	Started from 1999-01-04 until 2026-09-16
Number of rows	7093

Fill this document in as you work through the notebook. Each section matches one phase. Write in your own words — do not paste code. If a question asks "why", the answer is the part that matters.

The claim you are testing

"The euro has weakened against other major currencies since it launched."

Remember: both rates say how much one euro buys. When the number goes up, the euro is stronger.

Phase 1 — Getting the data

1.1 What did you load?

Question	Your answer
How many rows and columns did the full file have?	7094 row 43 columns
How many currency columns are there in total?	41
Which three columns did you keep?	Date, USD,CHF

1.2 Column types

One column was not the type you would expect. Which one, and why does that matter?

Date : object
USD :float64
CHF : flot64

1.3 Coverage

Question	Your answer
First date in the data	04/01/1999
Last date in the data	16/09/2026
Total number of rows	7094
Roughly how many rows per year?	around 253 days (you can explain your calculation)

Why is that number not 365?

There are fewer than 365 rows per year because exchange rates are not published on weekends and some public holidays.

Phase 2 — Cleaning

2.1 What you changed

List each cleaning step you carried out, and say why it was needed.

Step	What you did	Why it was needed
Date column	Converted the Date column to datetime format.	So pandas can treat the values as real dates and use them correctly for sorting and time-based analysis.
Row order	Sorted the rows by date from oldest to newest	Daily changes are calculated in the correct time order
Missing values	Checked columns of USD and CHF for missing values	To make sure there were no missing values that could effect on Data Analysis

2.2 The row order problem

The ECB publishes the newest row first. Explain, in your own words, what would go wrong if you calculated daily changes without sorting the data. Would you get an error message?
Without sorting the data, we would calculate the daily changes in the wrong order. It would be difficult to understand correctly if the exchange rate increased or decreased compared with the previous day.
We would not get an error message, but the result would be wrong.

2.3 Missing values

Question	Your answer
How many missing values in USD?	0
How many missing values in CHF?	0
Did you need to fill or drop anything?	No

Some currencies in the full file DO stop partway through. Name one, and say why it stopped.

RUB
The Russian Rouble (RUB) stopped in March 2022 because the ECB could no longer publish a fair exchange rate for it.

2.4 Does the data look sensible?

Write the smallest and largest value of each currency, and say whether they look believable.

Currency	Smallest	Largest	Believable?
EUR/USD	0.8252	1.599	yes
EUR/CHF	0.9008	1.6803	yes

Phase 3 — Exploring

3.1 Average rate by year

Write the average rate for five years spread across the period.

Year	Average EUR/USD	Average EUR/CHF
1999	1.0658	1.6003
2000	0.9236	1.5579
2001	0.8956	1.5105
2002	0.9456	1.4670
2003	1.1312	1.5212

What is the difference between how the two currencies behave over time?

EUR/USD moves up and down over time without a clear continuous trend. In contrast, EUR/CHF generally decreases over the long term, which shows that the euro has weakened more clearly against the Swiss franc.

3.2 Daily changes

Question	Your answer
Average daily change, EUR/USD	0.581
Average daily change, EUR/CHF	0.392
Which currency moves more from day to day?	USD

Why is the average daily change so close to zero, when the rates clearly do change over the years?

The average daily change is close to zero because positive and negative daily changes cancel each other out over time. The exchange rates still move a lot from day to day.

3.3 The biggest single day

Question	Your answer
Date of the largest single-day fall	15 January 2015
Which currency was it?	CHF
How large was the fall (%)?	14.4

What happened on that day? Write two or three sentences.

On 15 January 2015, the Swiss National Bank removed the minimum exchange rate of 1.20 CHF per euro.
The Swiss franc strengthened sharply, so the EUR/CHF rate fell significantly.

3.4 The quiet years

Look at how much EUR/CHF moved in 2012, 2013 and 2014, and compare it with 2015.	
Year	Typical daily change, EUR/CHF (%)
2012	0.105
2013	0.292
2014	0.111
2015	1.002

Those quiet years were not a sign that the franc was safe. Explain why.

The low volatility in 2012–2014 did not mean that a large movement was impossible. In 2015, EUR/CHF experienced a very large sudden change.

Phase 4 — Answering the question

4.1 First day to last day

Currency	First value	Last value	Change (%)
EUR/USD	1.18	1.15	-2.61
EUR/CHF	1.62	0.95	-41.45

On this evidence alone, does the claim hold? Answer in one sentence.

Yes, based on this evidence, the claim holds for both EUR/USD and EUR/CHF because both exchange rates decreased from the first day to the last day.

4.2 Starting from a different year

Copy eight rows from your table — pick a mix, including any year where the euro went UP against the dollar.

Starting year	EUR/USD change (%)	EUR/CHF change (%)
1999	-2.61	-41.45
2001	21.04	-37.80
2005	-15.00	-38.71
2010	-20.21	-36.55
2015	4.67	21.26
2020	2.57	-12.88
2023	7.47	-4.12
2025	11.24	1.01

4.3 Counting rises and falls

Question	Your answer
How many starting years did you test?	28
EUR/USD fell from how many of them?	16
EUR/CHF fell from how many of them?	25

4.4 Comparing the two fairly

Why can you not compare 1.08 dollars directly with 0.93 francs? What did you do instead?

We cannot compare the raw values directly because USD and CHF are different currencies. I indexed both rates to 100 so I could compare how much each one changed over time on the same scale.

4.5 Do the two move together?

Question	Your answer
Correlation of the two daily change columns	0.193
Closer to 0 or closer to 1?	0

What does that number mean for this project? Would one currency alone have been enough?

The correlation is low, so EUR/USD and EUR/CHF do not move closely together. One currency alone would not be enough to decide whether the euro has weakened.

Phase 4 — Your verdict

This is the most important page in the report. Answer each question in full sentences. There is no single right answer to the last one — what matters is that you can defend yours.

Does the claim hold for EUR/CHF?

Yes. The claim clearly holds for EUR/CHF. The euro weakened substantially against the Swiss franc over the period, with the rate falling by about 41%.

Does the claim hold for EUR/USD?

The claim only partly holds for EUR/USD. From the first day to the last day, the euro weakened slightly against the US dollar, but the result depends strongly on the starting year.

Which single piece of evidence decided it for you?

The starting-year comparison was the most useful evidence for me because it showed that EUR/CHF fell from most starting years, while the result for EUR/USD changed depending on the starting year.

Someone tells you "the euro has weakened". What two questions do you ask them, and why?

Against which currency?

Since when? / Over what time period?

I would also want to see the exchange rates at the start and end of the period.

Phase 5 — Charts

For each chart: tick it once it is finished, then write one sentence saying what it shows. A chart without a conclusion is decoration.

Chart	Done	What it shows (one sentence)
5.1 Two rates over time		EUR/USD fluctuates over time, while EUR/CHF shows a clearer long-term decline.
5.2 Both indexed to 100		Both exchange rates were indexed to 100 so their relative changes could be compared on the same scale.
5.3 Change by starting year		EUR/USD can rise or fall depending on the starting year.
5.4 Histogram of daily changes		Most daily EUR/USD changes are close to zero, while very large positive or negative changes are rare.
5.5 Movement per year		Volatility changes from year to year, and EUR/CHF shows an especially large increase in volatility in 2015.
5.6 Scatter of the two		The scatter plot shows a weak positive relationship between the daily changes of EUR/USD and EUR/CHF.

Chart checklist

Before you hand in, check every chart has all of these:

- ☐ A title that says what the chart shows, not just the column names
- ☐ A label on the x axis
- ☐ A label on the y axis, including the units
- ☐ A legend, if there is more than one line or colour
- ☐ Nothing cut off at the edges — use `plt.tight_layout()`

5.7 Exported file

Question	Your answer
File name	euro_analysis.xlsx
Sheet names	Yearly, Start_Years
Did you check it opens correctly?	Yes

Final summary

About 150 words, for someone who will not read your notebook. Include: the claim you tested, your answer for each currency, the one chart or table that convinced you, and the date range you used.

In this project, I tested the claim that the euro has weakened against other major currencies since it was launched. I used ECB exchange-rate data from 4 January 1999 to 16 September 2026 and focused on EUR/USD and EUR/CHF. From the first date to the last date, EUR/USD decreased by about 2.6%, so the euro weakened slightly against the US dollar. However, this result depends strongly on the starting year. EUR/CHF decreased by about 41.4%, showing a much clearer long-term weakening of the euro against the Swiss franc. The starting-year comparison was the most useful evidence for me. EUR/USD fell from 16 of 28 starting years, while EUR/CHF fell from 25 of 28. This shows that saying “the euro has weakened” is too general without saying against which currency and over which time period.

What this data cannot tell you

Write two things your analysis cannot answer, and why.

The data cannot tell us why the exchange rates changed, because it only contains the dates and exchange-rate values, not economic or political events.

The data cannot tell us whether the euro weakened against all currencies, because this analysis only focused on EUR/USD and EUR/CHF.

Marks are given for: cleaning decisions you can justify, charts a stranger could read, conclusions that match your own numbers, and honesty about the limits of your evidence. A careful "it depends, and here is why" beats a confident answer that the data does not support.